Shiv Nadar University, Chennai

Department of Information Technology

25IT1731 Computational Thinking and Problem Solving with Python

Ex 4 – String Handling

Part A

1	Determine the output of the following Python string operations: MyString = "Data Science 2026" a) print(MyString[0:4]) b) print(MyString[len(MyString)-1]) c) print(MyString[-6:-1]) d) print(2 * MyString) e) print(MyString[::-2]) f) print(MyString[:4] + MyString[4:]) g) print(MyString.swapcase()) h) print(MyString.startswith("Data") i) print(MyString.endswith("2026")) j) print(MyString.upper()) k) print(MyString.lower()) l) print(MyString.count("a")) m) print(MyString.find("Science")) n) print(MyString.index("Data")) o) print(MyString.replace("Science", "Analytics")) p) print(MyString.split(" ")) q) print(MyString.isalpha()) r) print("2026".isdigit()) s) print("Data2026".isalnum())
2	Write a Python program to read a string from the user and check whether the string is a palindrome.
3	Write a Python program to read a string and a character from the user and find the number of times the character occurs in the string without using the built-in count() method.
4	Write a Python program to read a string and count the number of: Uppercase letters, Lowercase letters, Digits, Special characters
5	Write a Python program to check whether a given password is strong. A password is considered strong if it contains: • At least 8 characters

	• At least one uppercase letter • At least one lowercase letter • At least one digit • At least one special character Display the requirements that are missing if the password is weak.
6	Write a Python program to read a string and replace every occurrence of character a with * without using the built-in replace() method. Input: banana Output: b*n*n*
7	Write a Python program to read a sentence and find: • Number of words • Longest word in the sentence Use appropriate built-in string methods.
8	Write a Python program for a university portal to validate an email address entered by a student. The email address should: • Remove leading and trailing spaces. • Be converted to lowercase. • End with one of the following domains: o @snuchennai.edu.in o @snu.edu.in
9	A document management system accepts only the following file types: .pdf,.docx.txt Write a Python program to read a filename and check whether it has an allowed extension.
10	Write a Python program that reads a social media post and displays all hashtags present in it. Input: Learning #Python and #AI at #SNUChennai Output: #Python #AI #SNUChennai

Part B

11	Write a Python program for a discussion forum that replaces a specified inappropriate word with *** Input: This is a bad message Word to replace: bad Output: This is a *** message. Do not use the built-in replace() method.
12	Write a Python program to compress a string by replacing consecutive repeated characters with the character followed by the number of times it occurs. Input: aaabbccccd Output: a3b2c4d1. Do not use buit in functions
13	**Course Registration Code Validation and Processing** A university course registration system uses course codes in the following format: **25IT1731** where the first two digits represent the year/batch, the next two characters represent the department code, and the last four digits represent the course number. Write a Python program to read a course code from the user and perform the following operations: 1. Remove any leading and trailing spaces from the input. 2. Convert the course code to uppercase. 3. Validate the course code based on the following conditions:

	- It must contain exactly 8 characters. - The first two characters must be digits. - The next two characters must be alphabets. - The last four characters must be digits. 4. If the course code is invalid, display: Invalid Course Code 5. If the course code is valid: - Display Valid Course Code. - Extract and display the year/batch, department code, and course number. - Display the message: Course registration details verified successfully. **Example:** Input: 25it1731 Output: Valid Course Code Year/Batch: 25 Department: IT Course Number: 1731 Course registration details verified successfully.

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